

מבחן 2007

מועד ד

חלק א

תכנון

```
library IEEE
;use IEEE.STD_LOGIC_1164.ALL

entity Address_Decoder is
    Port
        Address : in std_logic_vector(3 downto 0) -- כתובת כניסה בגודל 4 ביטים
        Detect : out std_logic -- פלט שמסמן זיהוי (Active High)
    ;
end Address_Decoder

architecture Behavioral of Address_Decoder is
    begin
        process(Address)
            begin
                if (Address = "0101" or Address = "1010") then
                    Detect <= '1'; -- Active High
                    התאמה נמצאה
                else
                    Detect <= '0'; -- אין התאמה
                end if
            end process
        end Behavioral
    end
```

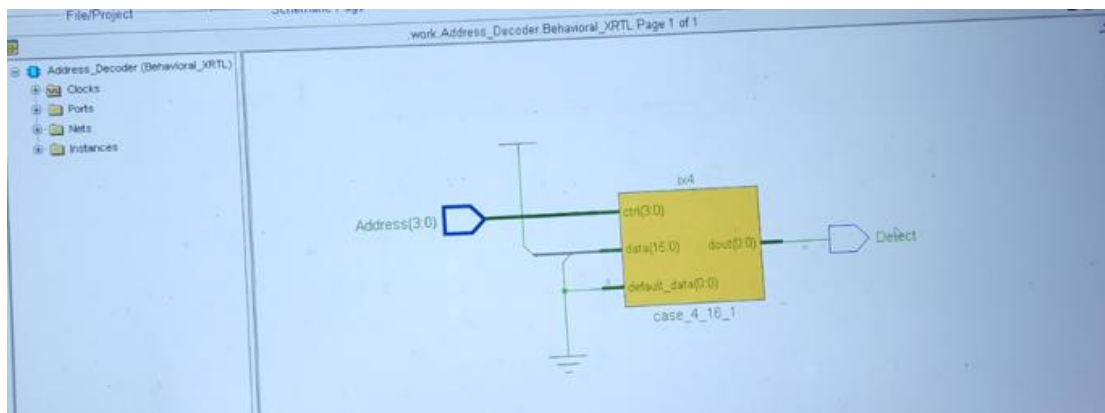
חלק ב

סינטזה

יוסי ברים
צוות 1

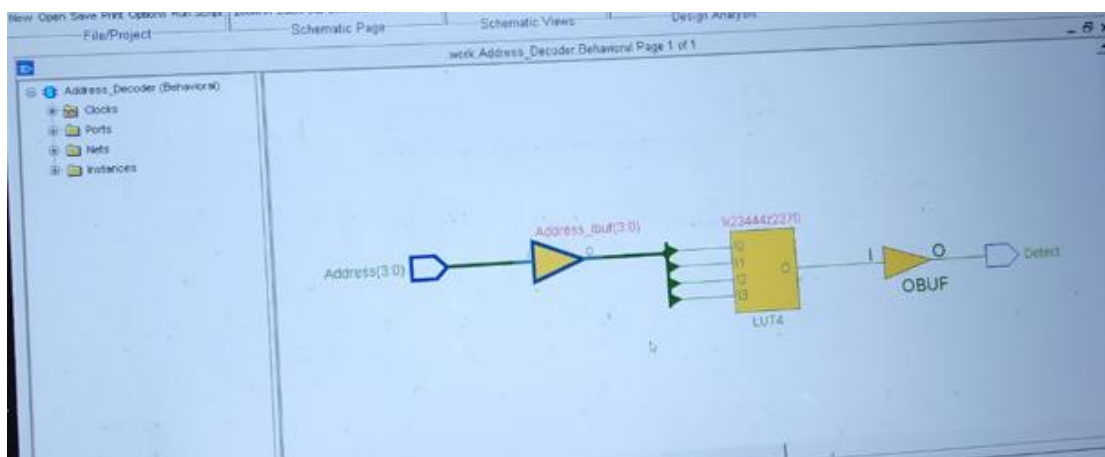
סעיף א

Rtl



סעיף ב

טכנולוגי



סעיף ג

Area report

Precision RTL Validate 64-bit 2018.1.0.19 (Production Release) Fri Aug 10 05:55:53 PDT //
2018

//

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//

Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //

//

Start time Sun Dec 15 17:58:03 2024 //

Device Utilization for 7A15TCPG236

Resource	Used	Avail	Utilization

IOs	5	106	4.72%
Global Buffers	0	32	0.00%
LUTs	1	10400	0.01%
CLB Slices	1	8150	0.01%
Dffs or Latches	0	20800	0.00%
Block RAMs	0	25	0.00%
DSP48E1s	0	45	0.00%

Library: work Cell: Address_Decoder View: Behavioral

Cell Library References Total Area

IBUF xca7 4 x

LUT4 xca7 1 x 1 1 LUTs

OBUF xca7 1 x

Number of ports : 5

Number of nets : 10

Number of instances : 6

Number of references to this view : 0

: Total accumulated area

Number of LUTs : 1

Number of accumulated instances : 6

IO Register Mapping Report

Design: work.Address_Decoder.Behavioral

Port	Direction	INFF	OUTFF	TRIFF
			Address(3)	Input
			Address(2)	Input
			Address(1)	Input
			Address(0)	Input
		Detect		Output

Total registers mapped: 0

סעיף ד

Timing report

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Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //

//

Start time Sun Dec 15 17:58:03 2024 //

Device: Xilinx - ARTIX-7 : 7A15TCPG236 : 3 --

..CTE report summary --

POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE RELIED ON TO --
MAKE QoR DECISIONS. For accurate timing information, please run place-and-route (P&R)
.and review P&R generated timing reports

=====

Clock Frequency Report

Domain	Clock Name	Min Period (Freq)	Required Period (Freq)
-----	-----	-----	-----
VirtualClock_PS	VirtualClock	4.348 (229.991 MHz)	10.000 (100.000 MHz)

=====

Setup Timing Analysis of VirtualClock

Setup Slack Path Summary

Data			Data			Data		
Setup Path			Data	End	End			
Index	Slack	Delay	Source Clock	Dest. Clock	Start Pin	Pin	Edge	

VirtualClock	VirtualClock	Address(3)	Detect	Rise	4.348	5.652	1	
VirtualClock	VirtualClock	Address(0)	Detect	Rise	4.348	5.652	2	
VirtualClock	VirtualClock	Address(1)	Detect	Rise	4.348	5.652	3	
VirtualClock	VirtualClock	Address(2)	Detect	Rise	4.348	5.652	4	

.Showing 4 paths found in the design

CTE Path Report

:Critical path #1, (path slack = 5.652)

SOURCE CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 1st rising edge

DEST CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 2nd rising edge

NAME	GATE	DELAY	ARRIVAL	DIR	FANOUT
	Address(3)	(port)		0.000	dn
	Address(3)	(net)	0.000		1
	Address_ibuf(3)/I	IBUF		0.000	dn
	Address_ibuf(3)/O	IBUF	1.147	1.147	dn

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צוות 1

Address_int(3)	(net)	0.295	1
ix23444z2370/I0	LUT4	1.442	dn
ix23444z2370/O	LUT4	0.097	1.539 up
nx23444z1	(net)	0.295	1
Detect_obuf/I	OBUF	1.834	up
Detect_obuf/O	OBUF	2.514	4.348 up
Detect	(net)	0.000	0
Detect	(port)	4.348	up

Initial edge separation: 10.000

Source clock delay: - 0.000

Dest clock delay: + 0.000

Edge separation: 10.000

Setup constraint: - 0.000

Data required time: 10.000

Data arrival time: - 4.348 (86.43% cell delay, 13.57% net delay)

Slack: 5.652

=====

Input Delay Report

Input	Clock Name	Slack (ns)
-----	-----	-----
Address(3)	VirtualClock	5.652
Address(2)	VirtualClock	5.652
	Address(1)	VirtualCloc

סעיף ה

סינטזה

synthesize

```
.Info: [662]: Current working directory: Z:/vhdlab6/project_1_impl_1 #  
Warning: [19539]: No debug point is implemented, continuing with the normal synthesis #  
.run  
  
Info: [15002]: Optimizing design <...>.Address_Decoder.Behavioral #  
Info: [15002]: Optimizing design <...>.Address_Decoder.Behavioral #  
  
...Info: [8010]: Gated clock transformations: Begin #  
...Info: [8010]: Gated clock transformations: End #  
  
Info: [1081]: Constraints: Precision will generate the following clock domain information #  
against the clocks. Since the clock domain information is not provided by the user for the  
below clocks, all clocks are assumed to be synchronous clocks. This will result in all paths  
between the clock domains to be analyzed for timing. If this behaviour is not intended for  
any clock domain or path, please override using -domain option or false path constraint.  
Among these clocks, clock with the smallest period will be chosen as the default domain and  
.the rest of the clock periods specified using multiply-by or divide-by factor  
  
:Info: Constraints: Precision Generated Clock Domains #  
  
Info: Domain Name: VirtualClock_PS      Clock Name: VirtualClock #  
  
.Info: [3027]: Writing file: <...>/project_1_impl_1/Address_Decoder.edf #  
  
Info: -- Writing file Z:/vhdlab6/project_1_impl_1/Address_Decoder.tcl #  
  
.Info: [3027]: Writing file: <...>/project_1_impl_1/Address_Decoder.xdc #  
  
.Info: [3027]: Writing file: <...>/project_1_impl_1/Address_Decoder.v #  
  
Info: -- Writing file Z:/vhdlab6/project_1_impl_1/Address_Decoder.tcl #  
  
...Info: [12045]: Starting timing reports generation #  
  
...Info: [12046]: Timing reports generation done #  
  
Info: [12048]: POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE #  
RELIED ON TO MAKE QoR DECISIONS. For accurate timing information, please run place-and-  
.route (P&R) and review P&R generated timing reports  
  
...Info: [665]: Finished synthesizing design #  
  
...Info: [11019]: Total CPU time for synthesis: 0.6 s secs #  
  
...Info: [11020]: Overall running time for synthesis: 2.3 s secs #
```